

PROBLEM A: Flood Planning

Lake Murray in central South Carolina is formed by a large earthen dam, which was completed in 1930 for power production. Model the flooding downstream in the event there is a catastrophic earthquake that breaches the dam. Two particular questions:

- Rawls Creek is a year-round stream that flows into the Saluda River a short distance downriver from the dam. How much flooding will occur in Rawls Creek from a dam failure, and how far back will it extend?
- Could the flood be so massive downstream that water would reach up to the S.C. State Capitol Building, which is on a hill overlooking the Congaree River?

MCM2005 A 题：洪水估计

位于美国南卡罗莱纳州中部的 Murray 湖是因为发电需要而建造的一座大型土坝而形成的，大坝建成于 1930 年。假设一场灾难性的大地震造成大坝决口，试对由此而产生的下游洪水进行建模。两个值得注意的问题：

- 1、Rawls 河是一条四季不断流的、距大坝下游很近的 Saluda 河的支流。当大坝决口形成洪水的时候，Rawls 河将遭受多大的洪水？河水倒流会延伸多远？
- 2、洪水会波及位于 Congaree 河边一座小山上的南卡罗莱纳州议会大厦吗？